Spatio-Temporal Feature Fusion and Guide Aggregation Network for Remote Sensing Change Detection

Hongguang Wei, Nan Wang, Yuan Liu, Pengge Ma, Dongdong Pang, Xiubao Sui, and Qian Chen

***Abstract*— The field of remote sensing change detection (RSCD) has seen significant advancements recently, focusing on the precise identification and analysis of temporal changes in remote sensing images. Existing deep learning-based RSCD methods primarily rely on concatenation or subtraction to integrate features of bi-temporal images and reconstruct change features through a feature pyramid network (FPN) decoding architecture. However, these methods face challenges related to inadequate spatio-temporal change representation and insufficient aggregation of multilevel semantic information, resulting in pseudo-changes and poor completeness of detected change objects. In this article, we propose an innovative RSCD framework via spatio-temporal feature fusion and guide aggregation (STFF-GA) to address the aforementioned challenges. The architecture of this network comprises two key components: the STFF module and the GA module. The STFF module is designed as a low-parameter and low-computation structure, effectively enhancing the representation of spatio-temporal change information through split, interaction, and fusion strategies. The GA module uses deep feature guidance (DFG) mapping as prior information to guide the aggregation of multilevel semantic information, thereby correcting the positional information of change objects and filtering out pseudo-changes and other noise interference. In addition, it utilizes convolution kernels of various scales to extract fine-grained features, facilitating the complete reconstruction of change objects. Extensive experiments conducted on three benchmark change detection datasets demonstrate that the proposed STFF-GA consistently outperforms other state-of-the-art (SOTA) detectors. The code is available at https://github.com/NjustHGWei/STFF-GA.**

***Index Terms*— Change detection, convolutional neural network (CNN), guide aggregation (GA), spatio-temporal feature fusion (STFF).**

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I. Introduction

REMOTE sensing change detection (RSCD) aims to analyze and detect variations in objects of interest between different periods within the same geographic region. It provides valuable information for understanding the dynamics on the surface of the Earth. The RSCD technology has been widely applied in various scenarios, including building change detection, vegetation change detection, water body change detection, road change detection, and so on [1], [2]. By detecting and tracking these changes, important geographic information and environmental monitoring data can be provided. These data are crucial for applications such as urban planning [3], land and resources management [4], and disaster monitoring [5].

Traditional methods of change detection typically require manual selection and extraction of features. The specific detection process can be summarized as follows: initially, bi-temporal remote sensing images undergo preprocessing, which includes techniques like image registration and radiometric correction [6]. Subsequently, the distinction between changed and unchanged pixels is established through methods such as image differencing [7] and image ratioing [8]. Finally, the change difference features are processed to generate a change result map, which typically involves techniques such as principal component analysis [9] and threshold segmentation. However, this process relies on manually extracted features, rendering it susceptible to various disturbances, such as noise, alignment discrepancies, and illumination variations. Afterward, machine learning techniques gradually emerged, utilizing algorithms like decision trees [10], k-nearest neighbors [11], and random forest regression [12] to construct decision models for the binary classification of pixels. Although the introduction of learning-based methods in the decision-making phase has somewhat enhanced accuracy, robustness remains limited.

Compared to traditional methods, deep learning-based approaches demonstrate superior detection performance and

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generalization capabilities. These approaches can be categorized based on the techniques employed in their model encoders and decoders: convolutional neural network (CNN)-based methods, transformer-based methods, and hybrid CNN-transformer methods. CNN-based change detection methods refer to models that primarily use CNN for feature encoding and decoding, which excels in local feature extraction. For example, Daudt et al. [13] first proposed three different structures for change detection based on the UNet, including the FC-EF network, FC-Siam-conc network, and FC-Siam-diff network. These methods explore different input modalities and fusion strategies for bi-temporal images. Further advancements have focused on enhancing the extraction of change information by incorporating attention mechanisms [14], [15], [16], [17]. Researchers have repeatedly attempted to use these mechanisms to adjust CNN models, thereby improving their focus on change features. However, the inherent limitations of the local receptive fields in CNNs hinder their ability to capture global information. To address this, more network layers are typically stacked to expand the receptive field, which significantly increases the number of parameters and computational complexity. Subsequently, several studies have incorporated self-attention mechanisms [18], [19], [20] into CNN models, which has alleviated the limitations of the receptive fields in CNN models to some extent.

Transformer-based change detection methods apply transformer techniques in the encoder or decoder, excelling at capturing long-range dependencies and global information. These methods are well demonstrated in networks such as bi-temporal image transformer (BIT) [21], ChangeFormer [22], and RSP-BIT [23]. In addition, some studies have proposed hybrid change detection methods that combine the strengths of CNN and transformer technologies. These methods typically involve using separate CNN and transformer branches during the encoding stage to extract bi-temporal change features, followed by decoding to reconstruct the feature space representing change information [24], [25]. For instance, Feng et al. [26] proposed an intrascale cross-interaction and interscale feature fusion network (ICIF-Net), which synergizes CNN and transformer to capture the local–global features of multitemporal remote sensing images concurrently. Tang et al. [27] developed a W-shaped hierarchical network, termed WNet, which facilitates change detection by progressively amalgamating features from CNN and transformer layers during the decoding phase. Although methods utilizing transformer technology can effectively capture global change features and enhance change detection performance, most transformer-based models process high-resolution remote sensing images by dividing them into patches. The division of these patch blocks disrupts the continuity and geometry of the object of interest in the high-resolution remote sensing image, thus adversely affecting the change detection task.

In summary, while CNN-based, transformer-based, and hybrid CNN-transformer methods have achieved effective performance in change detection tasks, these methods are typically used as feature extractors for single-temporal images, and they lack exploration in fusing bi-temporal remote sensing image features. Current bi-temporal feature fusion methods commonly use concatenation or subtraction [28], [29] to integrate change features. However, these approaches are inefficient in representing change information, and there is still much room for improving their performance.

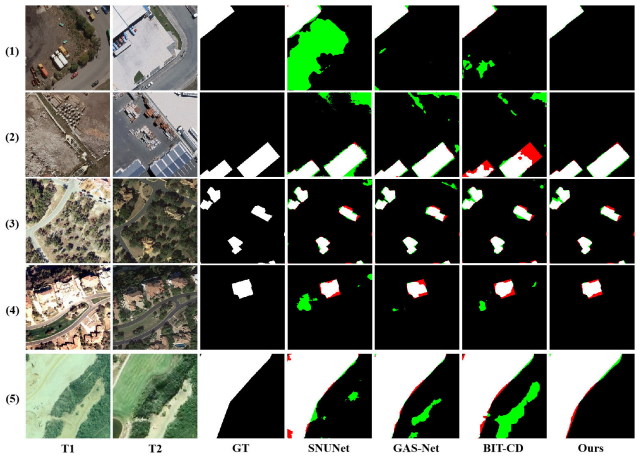


Fig. 1. Examples of some typical pseudo-change situations. Red areas indicate missed detection, while green areas indicate erroneous detection. In rows (1 and 2), pseudo-changes appear in ground regions due to changes in non-interesting objects. In rows (3–5), pseudo-changes result from inconsistencies in season or illumination.

In addition to bi-temporal feature fusion, the aggregation of multilevel semantic information plays a crucial role in accurately identifying change objects and reconstructing their completeness [30]. As shown in Fig. 1, higher image resolution enriches the semantic information contained in images. However, it also introduces more interference factors. For instance, changes of noninteresting objects in (1) and (2), and color differences caused by varying lighting or seasonal conditions in (3)–(5). Besides, some advanced models in Fig. 1 struggle to capture complete change objects, while certain pseudo-change regions are incorrectly identified as real changes. These interference factors challenge the aggregation of multilevel semantic information, making it difficult to reconstruct change objects accurately [31]. Current methods typically employ feature pyramid network (FPN) [32] architectures for multilevel semantic information aggregation [33], [34]. However, these methods do not adequately consider the semantic gap between features at different levels, leading to inaccurate detection results [35]. Therefore, exploring more effective methods to aggregate multilevel semantic information is necessary, reducing the impact of pseudo-changes and other interference factors, and facilitating the integrity reconstruction of change objects.

To address the aforementioned issues, we propose a spatio-temporal feature fusion and guide aggregation (STFF-GA) network. This network uses a weight-sharing Siamese ResNet18 as the backbone to extract multilevel features, ranging from coarse to fine, from bi-temporal remote sensing images. Given the inefficiency of fusing bi-temporal features through concatenation or subtraction, we designed a spatio-temporal feature fusion (STFF) module, which employs a strategy of splitting, interaction, and fusion to integrate bi-temporal features, enhancing the representation of change information. Specifically, the STFF module first obtains a

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difference map of the bi-temporal features through subtraction operations, i.e., coarse change features. The subtraction operation focuses on the spatial dimension, directly reflecting the degree of change between the bi-temporal features, which is beneficial for subsequent processing. The splitting and interaction operations allow us to divide the bi-temporal and coarse change features into different components, facilitating fine-grained feature interaction and strengthening the information flow between them. The feature interaction employs addition operations aimed at enhancing their common features while retaining their unique information, thereby highlighting areas that contribute to the integrated representation. Moreover, different levels of change features exhibit semantic gaps, and lower level features may contain pseudo-changes and other noise. Directly connecting different levels of features without proper processing can reduce the representation capacity of multiscale change features and allow noise in lower level features to contaminate the change information, leading to inaccurate detection results. Therefore, we developed a guide aggregation (GA) module that utilizes deep feature guidance (DFG) mappings as prior information to guide the aggregation of multilevel semantic information, thereby correcting the positional information of change objects and filtering pseudo-changes and other background noise. Subsequently, multiscale convolution operations are used to extract fine-grained change features, promoting the complete reconstruction of change features.

The main contributions of this article can be summarized as follows.

1) We propose the STFF-GA, an innovative change detection framework that leverages the STFF and GA modules to localize change objects precisely. Experimental results on three datasets demonstrate that STFF-GA exhibits superior detection performance and robustness.
2) The STFF module is designed as a low-parameter, low-computation module. It integrates bi-temporal features through a strategy of splitting, interaction, and fusion, thereby enhancing the representation of spatio-temporal change features. This provides crucial detection cues for the subsequent reconstruction of change objects.
3) The developed GA module integrates the DFG module and multiscale convolution operation, which addresses the issue of insufficient aggregation of multilevel semantic information, effectively filters the interference of pseudo-change and other background noises, and facilitates the integrity reconstruction of changed objects.

II. Related Work

Existing CNN-based change detection methodologies predominantly employ the Siamese UNet encoding–decoding framework. Notably, the same level of features in the dual streams need to be fused in the feature extraction stage to obtain the change features, which is different from the traditional semantic segmentation task. In the sequel, we succinctly revisit the Siamese UNet-based change detection model and the bi-temporal feature fusion strategy, both of which constitute the foundational research of our work.

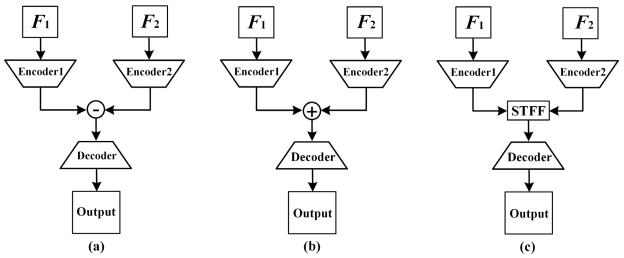


Fig. 2. Bi-temporal feature fusion structure. (a) Element-wise subtraction. (b) Channel-wise concatenation. (c) STFF.

A. Siamese UNet Architecture-Based CD Methods

In recent years, the traditional UNet was extended, and many pioneering researches were presented by considering specific scenarios and requirements of change detection tasks. Lv et al. [36] enhanced the Siamese UNet architecture with multiscale convolutional modules, enabling the refined detection of change objects across various sizes and shapes within image scenes. Sun et al. [37] innovated with a spatio-temporal network, replacing convolutional layers of UNet with long short-term memory convolutions to leverage both temporal and spatial dimensions of bi-temporal images. Subsequently, various variant structures have emerged from the Siamese UNet network. For instance, Peng et al. [38] utilized the UNet++ model for an end-to-end change detection approach, employing a multiside output fusion strategy for generating precise change maps. Fang et al. [28] proposed a novel change detection network integrating the Siamese network and nestedUNet (SNUNet). Miao et al. [39] proposed a full-scale connected UNet (SNUNet3+) integrated network. These networks get more fine-grained change information through variants of the UNet architecture.

In addition, Liu et al. [40] and Li et al. [29] replaced the Siamese UNet backbone to realize a lightweight change detection network with a lightweight shufflenetv2 [41] model and MobileNetV2 [42] model, respectively. These architectures, while lightweight, face challenges in capturing complex spatio-temporal change features due to their reduced parameters and simplified structures, complicating the balance between efficiency and accuracy. Conversely, Han et al. [43] and Zhou et al. [44], respectively, replaced the backbone network of Siamese UNet with VGG [45] and ResNet [46] models, achieving fine-grained extraction of change features, thereby improving the performance of change detection. The pretrained models are used for training change detection tasks, which can accelerate convergence speed and improve training efficiency. Consequently, this article utilizes pretrained ResNet18 as the backbone feature extraction network to achieve fine-grained extraction of change features and accelerate model convergence during training.

B. Bi-Temporal Feature Fusion Network

The existing change detection methods largely extend from semantic segmentation techniques. However, in change detection tasks, multilevel features are extracted usually using two-stream networks. To optimally amalgamate change features

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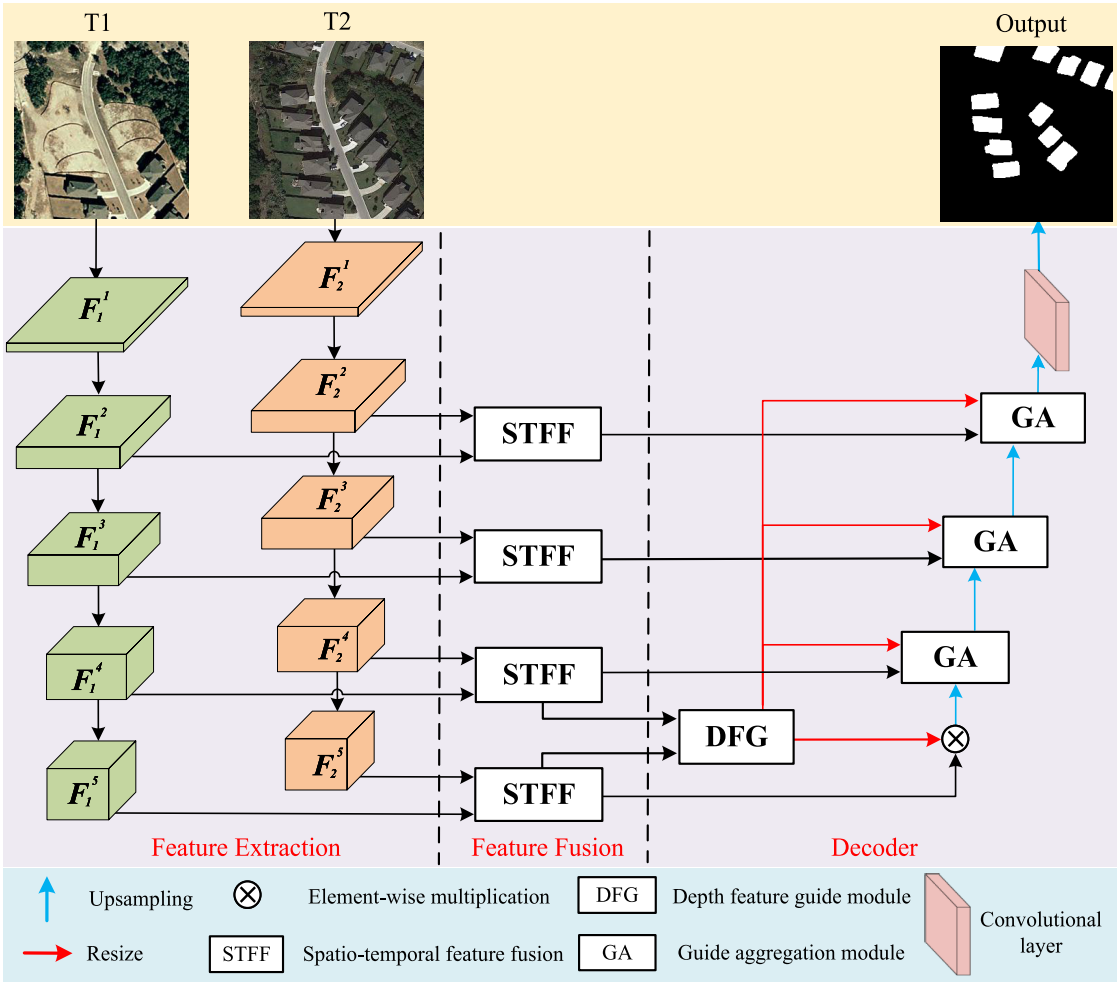


Fig. 3. Overall architecture of the STFF-GA network.

from bi-temporal images, several studies have underscored the efficacy of bi-temporal feature fusion techniques [47], [48]. The traditional methods for fusing bi-temporal features usually involve simple element-wise subtraction [as shown in Fig. 2(a)] or channel-wise concatenation [as shown in Fig. 2(b)]. However, both of these methods exhibit poor adaptability to change detection techniques.

Subsequently, researchers have proposed a series of novel bi-temporal feature fusion techniques. For example, Ma et al. [49] introduced a temporal feature fusion module utilizing a cross-time gating mechanism to selectively accentuate relevant changes while suppressing irrelevant changes. Similarly, Zhao et al. [50] developed a temporal fusion module aimed at pinpointing crucial elements across bi-temporal features for their effective integration. Wang et al. [51] proposed a feature interaction fusion module designed to exploit shared and differential information between bi-temporal features, thereby generating distinguishable fused features. These methods integrate advanced bi-temporal fusion architectures incorporating attention mechanisms and convolutional operations [52], [53], enhancing the modeling capability for the interactive fusion of bi-temporal features. However, they also correspondingly increase the parameter and computational complexity of the model. Hence, this article proposes a low-parameter and low-computation STFF module [as shown in Fig. 2(c)] to enhance the interactive fusion of bi-temporal features.

III. Methodology

A. Overall Architecture

The proposed STFF-GA network architecture adopts a Siamese structure and follows an end-to-end U-shaped encoding–decoding style. Fig. 3 illustrates the overall structure of the STFF-GA network, which includes a standard Siamese ResNet18 model with shared weights as the backbone, an STFF module for fusing bi-temporal features, and the decoding section with DFG and GA modules.

The implementation steps for the proposed STFF-GA network in remote sensing image change detection can be described as follows: initially, a pair of registration remote sensing images (marked as T1 and T2) captured at different times is fed into the Siamese ResNet18 encoder to extract multilevel features spanning from coarse to fine. The extracted multilevel features are denoted as F_1^i and F_2^i, $i \in \{1, 2, 3, 4, 5\}$. Here, we utilize the last four feature layers for decoding, with spatial sizes of 64×64 pixels, 32×32 pixels, 16×16 pixels, and 8×8 pixels, respectively. Subsequently, a low-parameter and low-computation STFF module enables

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the interactive fusion of bi-temporal features, significantly bolstering the ability of the model to capture the change features and augment the representation of change information. Following this, the deep high-level semantic information (F^4_{STFF} and F^5_{STFF}) and self-attention mechanism are utilized to generate a DFG mapping with global semantic information. The DFG mapping is integrated into the GA module as prior information to guide the aggregation of multilevel semantic information, which enables the correction of the positional information of the changing objects and filtering of pseudo-changes and other noisy interference. Concurrently, the GA module employs multiscale convolution operations to describe fine-grained change features, thereby enhancing the integrity reconstruction of change objects. The culmination of this process employs a convolution module to implement binary classification of changing and unchanged pixels.

B. STFF Model

Bi-temporal feature fusion is crucial for change detection architectures. It correlates bi-temporal images at the same feature level, maximizing the distinction and extraction of integrated feature changes. Therefore, the effectiveness of bi-temporal feature fusion methods largely determines the performance of change detection.

Previous methods have primarily relied on concatenation or subtraction to fuse temporal features. Concatenation integrates bi-temporal feature maps into a higher dimensional feature map, encapsulating all information from bi-temporal features. However, this high-dimensional feature map requires more parameters and computations for subsequent processing. On the other hand, subtraction focuses on the spatial dimension, producing a difference map that directly reflects the degree of change between the bi-temporal feature maps [54]. The greater the difference, the more significant the change. Typically, after subtraction, absolute value computation is needed to eliminate directionality, focusing only on the magnitude of change and retaining all change information. Although subtraction is relatively simple, effective, and computationally efficient, it is highly sensitive and may misinterpret pseudo-changes and other noise as change information, thereby affecting detection performance. In summary, while simple concatenation or subtraction methods for fusing bi-temporal features can enhance change detection performance, they have limitations. There is still a need to explore more effective bi-temporal feature fusion methods.

Based on concatenation and subtraction operations, we propose a novel STFF module to integrate change information and enhance its representation capabilities. The STFF module has the advantages of low parameters and low computational complexity. The architecture of the STFF module is shown in Fig. 4. Initially, we compute the difference feature map, denoted as $F^i_d \in R^{C_i \times H_i \times W_i}$, by an element-wise subtraction operation. The subtraction operation has been proven effective in several studies [29]. Here, i represents the ith level feature, and C_i, H_i, and W_i represent the channel count, height, and width of the F^i_d, respectively. Compared to concatenation, subtraction can more intuitively and effectively highlight changes, potentially providing clearer indications of change presence. The mathematical expression for this operation can be written as

$$F^i_d = \left|F^i_1 - F^i_2\right|, \quad i \in \{2, 3, 4, 5\}. \tag{1}$$

Inspired by [52] and [55], we further designed a bi-temporal change feature reconstruction method. Instead of directly concatenating the bi-temporal features and F^i_d, we employed a strategy of splitting, interaction, and fusion to reconstruct change features. This approach fully integrates three different types of information features (F_1, F_2, and F^i_d), enhancing the information flow between them. First, F_1, F_2, and F^i_d are each split along the channel dimension into two feature blocks, each feature block has a channel count of $C_i/2$. The splitting operation allows us to decompose bi-temporal features and F^i_d into different components, facilitating fine-grained processing. Next, we perform element-wise interaction and summation of the split features. The interaction and summation operation runs along the channel dimension, enhancing their common features while preserving their unique information [54]. This approach highlights the regions within the two features that contribute to the integrated representation. Formally, it's written as

$$\begin{cases} F^i_{1_1}, F^i_{1_2} = \text{split}\left(F^i_1\right) \\ F^i_{d_1}, F^i_{d_2} = \text{split}\left(F^i_d\right) \\ F^i_{2_1}, F^i_{2_2} = \text{split}\left(F^i_2\right) \end{cases} \tag{2}$$

$$\begin{cases} F^i_{\text{sum}_1} = F^i_{1_1} \oplus F^i_{d_2} \\ F^i_{\text{sum}_d} = F^i_{d_1} \oplus F^i_{2_2} \\ F^i_{\text{sum}_2} = F^i_{2_1} \oplus F^i_{1_2} \end{cases} \tag{3}$$

where $\oplus$ represents element-wise summation. Finally, to further enhance the representation capability of the change features, depthwise separable convolutions (DSCs) are applied to F_{sum_1}, F_{sum_2}, and F_{sum_d}. Subsequently, these features are concatenated into a higher dimensional feature map, which contains all the spatio-temporal change information extracted through the interaction and fusion of the bi-temporal features. Formally, it's written as

$$\begin{cases} F^i_{\text{DSC}_1} = \text{DSC}\left(F^i_{\text{sum}_1}\right) \\ F^i_{\text{DSC}_d} = \text{DSC}\left(F^i_{\text{sum}_d}\right) \\ F^i_{\text{DSC}_2} = \text{DSC}\left(F^i_{\text{sum}_2}\right) \end{cases} \tag{4}$$

$$F^i_{\text{STFF}} = \text{Cat}\left(F^i_{\text{DSC}_1}, F^i_{\text{DSC}_d}, F^i_{\text{DSC}_2}\right) \tag{5}$$

where $\text{DSC}(\cdot)$ denotes depthwise separable convolution. $\text{Cat}(\cdot)$ is a feature concatenation operation.

Notably, both the splitting and interaction steps do not introduce any additional parameters. Furthermore, the fusion step employs DSC, ensuring that the proposed STFF module maintains a low parameter count and computational complexity. By incorporating the STFF module into the framework, we achieve effective interaction and fusion of bi-temporal features without adding excessive computational complexity or parameter overhead, thereby enhancing the representation capability of spatio-temporal change features.

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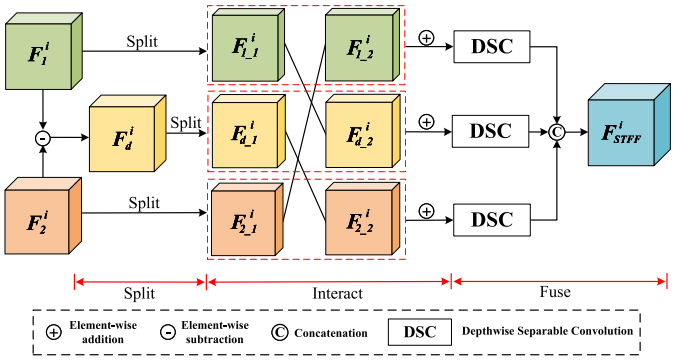


Fig. 4. Illustration of the STFF module.

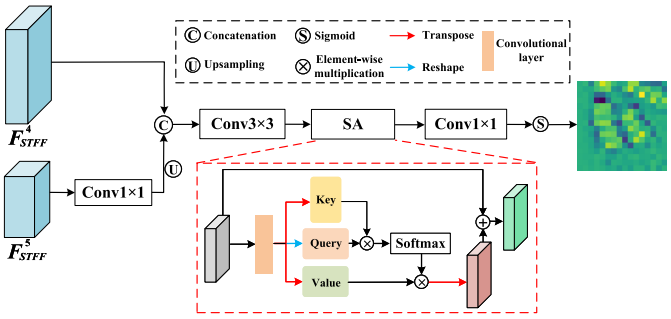


Fig. 5. Illustration of the deep feature guide module.

C. Change Feature Decode

Traditional FPN decoding architectures often overlook significant semantic gaps between different hierarchical features and the interference of background noises in lower level features [35]. Directly aggregating multiscale semantic information without any processing will reduce the representation ability of multilevel features, and may also introduce pseudo-changes and other noise interference in the detection results. To mitigate these challenges, some studies have employed guided mapping generated from different hierarchical features to aid in the aggregation of multilevel semantic information [43], [56]. Through empirical evaluations, we observe that guide mapping plays a crucial role in accurately locating change objects and filtering out pseudo-changes and other noise. However, the above two methods do not consider the problems that shallow features may contain pseudo-change and other background noises and high-level features with a high degree of abstraction may lead to the loss of detailed information. Moreover, relying solely on convolution operations makes it difficult to capture long-range dependencies and global information representations of changing objects, thereby affecting the accuracy of the detection results. Therefore, generating an accurate change-guided mapping remains a significant challenge.

In previous studies, it has been widely accepted that there exists a semantic gap between low-level and high-level features [57]. Specifically, low-level features contain rich spatial information such as edges, texture details, and irrelevant background noises, while high-level features contain abundant semantic information that aids in understanding the changed regions in images [51]. Moreover, high-level features are less sensitive to noise interference, demonstrating better stability and robustness. Based on these considerations, we integrate high-level features from scales F^4_{STFF} and F^5_{STFF}, along with a self-attention mechanism, to construct DFG mappings. In subsequent work, we will conduct ablation studies to elucidate the superiority of choosing F^4_{STFF} and F^5_{STFF} for generating DFG mappings. Our motivation for this design includes: 1) leveraging deep high-level features from two different scales to provide richer change and detailed information; 2) capturing global information representations of change features through self-attention mechanisms; and 3) avoiding interference from background noises in shallow-level features.

During the change feature decoding stage, the DFG mapping is embedded into the GA module to assign weight coefficients for different positions of the feature map. This process helps correct the positional information of change objects and filters out pseudo-changes and other background noise, thereby facilitating the aggregation of multilevel semantic information. Subsequently, multiscale convolution operations are utilized to extract fine-grained change features, enhancing the integrity reconstruction of change objects.

1) Deep Feature Guide Module: The DFG mapping provides valuable guidance for the subsequent GA module. The proposed DFG module generates the DFG mapping using two different scales of high-level semantic information and self-attention mechanism, assisted by deep supervision loss (DSL) to approximate the true change objects for more accurate guidance. Our design of the DFG mapping offers a more effective global information representation compared to methods in [43] and [56]. Subsequent ablation studies further emphasize the superiority of our approach.

As shown in Fig. 5, the input features F^4_{STFF} and F^5_{STFF} initiate the workflow of the DFG module. Considering the different spatial and channel dimensions of F^4_{STFF} and F^5_{STFF}, we use upsampling and 1×1 convolution to align the spatial scales and compress the channel dimensions of the high-level features. Subsequently, the two features at the same spatial scale are concatenated and further processed with a 3×3 convolution operation to enhance the feature representation. The formal process of the convolution operation is as follows:

$$F_{\text{Conv}} = \text{Conv}_{3\times3}\left(\text{Cat}\left(\left[F^4_{\text{STFF}}, \text{Up}\left(\text{Conv}_{1\times1}\left(F^5_{\text{STFF}}\right)\right)\right]\right)\right) \quad (6)$$

where $\text{Up}(\cdot)$ signifies upsampling operation, $\text{Conv}_{1\times1}$ denotes a 1×1 convolution, and $\text{Conv}_{3\times3}$ represents a 3×3 convolution.

Due to the limited receptive field of the 3×3 convolution kernel, the output is only partially related to the input, lacking sufficient global context information. Therefore, in subsequent processing, we introduce a self-attention mechanism, which plays a crucial role in enhancing the model's global information perception and long-term dependency capture. Moreover, the self-attention mechanism allows the model to selectively focus on the relevant parts of the input features, enhancing the attention to task-related information while minimizing the influence of irrelevant details. Next, we use a 1×1 convolution to compress the number of channels to 1, aiming to fuse the multidimensional feature information into a compact representation for subsequent processing. Finally, a Sigmoid

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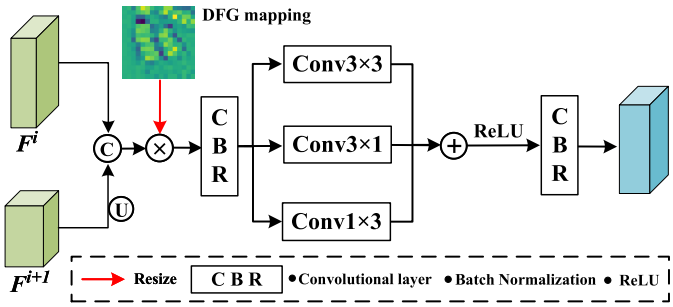


Fig. 6. Illustration of the GA module.

function is employed to generate the DFG mapping. The expression for the DFG mapping is as follows:

$$F_{\mathrm{DFG}} = \sigma(\mathrm{Conv}_{1\times1}(\mathrm{SA}(F_{\mathrm{Conv}}))) \tag{7}$$

where $\mathrm{SA}(\cdot)$ denotes the self-attention mechanism, $\sigma(\cdot)$ represents the Sigmoid function, and F_{DFG} denotes the output DFG mapping. This process ensures that the DFG mapping effectively integrates both local and global feature information, enhancing the representation performance of DFG mapping.

2) Guide Aggregation (GA) Module: Fig. 6 illustrates the workflow of the GA module. We use DFG mapping to guide the aggregation of multilevel semantic information. To achieve this, we first align the spatial scales of two different levels of features and then concatenate them along the channel dimension. Subsequently, the DFG mapping is used as prior information to impose weights on the concatenated multilevel features, this operation has several advantages: 1) mitigates the semantic gap between features at different levels; 2) positions containing changes are given greater weight, while the background is given smaller weight, reducing interference from pseudo-changes and other background noise; and 3) corrects the location information of change objects. The process is mathematically represented as follows:

$$F^i_{\mathrm{Cat}} = \mathrm{Cat}\left(\left[F^i_{\mathrm{STFF}}, \mathrm{Up}\left(F^{i+1}_{\mathrm{STFF}}\right)\right]\right) \otimes F_{\mathrm{DFG}} \tag{8}$$

where $\otimes$ signifies element-wise multiplication. Next, we apply a 1×1 convolution block to compress the number of channels. This helps reduce the model's computational complexity and combines the rich information from the high-dimensional feature space into a more compact form. The convolution formula can be expressed as

$$F^i_{\mathrm{CBR}} = \mathrm{CBR}_{1\times1}\left(F^i_{\mathrm{Cat}}\right) \tag{9}$$

where $\mathrm{CBR}_{1\times1}(\cdot)$ denotes the combination of a 1×1 convolution, batch normalization, and ReLU activation function.

In subsequent processing, we consider the poor integrity of the change objects. Inspired by [58], we utilize a series of convolution kernels of different scales, sizes of 3×1, 3×3, and 1×3, respectively, to extract fine-grained multiscale change features. Specifically, the 3×1 kernel focuses on vertical convolution for vertical change information extraction, whereas the 1×3 kernel specializes in horizontal convolution for horizontal change information extraction. The multiscale change information is then fused by summing and averaging. This architecture effectively captures fine-grained change features at different spatial scales, thereby enhancing the complete reconstruction of spatial details within the changed regions. The convolution formula can be expressed as follows:

$$F^i_{\mathrm{Conv}} = \left(\left[\mathrm{Conv}_{3\times3}\left(F^i_{\mathrm{CBR}}\right) \oplus \mathrm{Conv}_{3\times1}\left(F^i_{\mathrm{CBR}}\right) \oplus \mathrm{Conv}_{1\times3}\left(F^i_{\mathrm{CBR}}\right)\right]/3\right) \tag{10}$$

where $\mathrm{Conv}_{3\times3}(\cdot)$, $\mathrm{Conv}_{1\times3}(\cdot)$, and $\mathrm{Conv}_{3\times1}(\cdot)$ represent 3×3 convolution, 1×3 convolution, and 3×1 convolution, respectively.

Finally, the ReLU activation function and 1×1 convolutional blocks are used to promote the learning and fitting ability of the network. The mathematical style can be denoted as

$$F^i_{\mathrm{GA}} = \mathrm{CBR}_{1\times1}\left(\mathrm{ReLU}\left(F^i_{\mathrm{Conv}}\right)\right) \tag{11}$$

where $\mathrm{ReLU}(\cdot)$ represents the ReLU activation function.

D. Loss Function

For the RSCD tasks, the primary objective is to accurately classify pixels as either changed or unchanged. However, there is an extreme class imbalance between changed and unchanged pixels, which can cause the model to be biased toward predicting the dominant unchanged class while ignoring the minority changed class. To mitigate this imbalance, we employ a hybrid loss function that integrates binary cross-entropy (BCE) and dice loss, described by the following equation:

$$\mathrm{Loss} = \mathrm{Loss}_{\mathrm{BCE}} + \mathrm{Loss}_{\mathrm{dice}}. \tag{12}$$

BCE loss is represented as

$$\mathrm{Loss}_{\mathrm{BCE}}(t, p) = \frac{1}{N}\sum_{i=1}^{N}[t_i \log(p_i) + (1 - t_i)\log(1 - p_i)]. \tag{13}$$

Dice loss is represented as

$$\mathrm{Loss}_{\mathrm{dice}} = 1 - \frac{2 \cdot \sum_{i=1}^{N} t_i \cdot p_i}{\sum_{i=1}^{N} t_i + \sum_{i=1}^{N} p_i} \tag{14}$$

where N denotes the total pixel count, while t and p represent the ground truth labels and the predicted result map, respectively. For each pixel i, $t_i \in \{0, 1\}$ signifies the ground truth class, with "0" for unchanged and "1" for change. The predicted probability of pixel i being classified as changed is $p_i \in [0, 1]$ and $1 - p_i$ as unchanged.

The computation of the overall loss encompasses the prediction loss and the loss associated with the DFG mapping. Given that DFG mapping yields critical insights into change regions and augments the reconstruction capabilities of the decoder. Consequently, deep supervision of DFG mapping by calculating the loss between DFG mapping and ground truth labels can provide more accurate guidance of change object location information for the aggregation of multilevel semantic information, thereby improving the accuracy of change detection.

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IV. Experiments

In this section, we provide a comprehensive description of the datasets and comparison methods utilized in the experiments. Following this, the experimental framework is outlined, encompassing evaluation metrics and implementation specifics. Subsequently, a suite of experiments is conducted to evaluate the performance of our proposed STFF-GA network. These experiments analyze results through quantitative metrics, qualitative assessments, and efficiency comparisons. Finally, ablation studies are conducted to scrutinize the individual contributions of each module within the proposed method.

A. Dataset Description

1) *LEVIR-CD Dataset [59]:* Released by Beihang University, the LEVIR-CD is a high-resolution dataset dedicated to building change detection. It contains 637 bi-temporal aerial image pairs from 20 different regions in the United States, spanning temporal intervals of 5–14 years. Each image has dimensions of 1024×1024 pixels and a spatial resolution of 0.5 m. Due to GPU memory constraints, the dataset has been divided into nonoverlapping patches of 256×256 pixels. It is randomly split into training (7120 pairs), validation (1024 pairs), and testing sets (2048 pairs).
2) *WHU-CD Dataset [60]:* The WHU-CD dataset comprises portions of New Zealand, collecting bi-temporal remote sensing images captured after the earthquakes in 2011 and after the reconstruction in 2016. It features an ultrahigh-resolution image pair of $32\,507 \times 15\,354$ pixels. For practicality in training, the images have been segmented into 256×256 pixel patches. Lacking a separate validation dataset, 10% of the training image pairs were randomly selected to form a validation dataset, resulting in 4536 pairs for training, 504 pairs for validation, and 2760 pairs for testing.
3) *SYSU-CD Dataset [61]:* Released by Sun Yat-sen University, the SYSU-CD dataset includes 20 000 image pairs from the Hong Kong region, captured between 2007 and 2014. These images, with a spatial resolution of 0.5 m and 256×256 pixel dimensions, cover a variety of change types such as harbors, vegetation, and roads, offering a broader challenge in change detection tasks. The dataset is partitioned according to the SYSU-CD standard: 12 000 pairs for training, 4000 for validation, and 4000 for testing.

B. Compared Methods

To thoroughly evaluate the performance of our proposed algorithm, we compare it against nine state-of-the-art (SOTA) RSCD models. These include CNN-based models such as FC-EF [13], FC-Siam-conc [13], FC-Siam-diff [13], SNUNet [28], DMINet [20], and GASNet [18], alongside transformer-based approaches like BIT [21]. In addition, hybrid models integrating CNNs with transformers, namely, ICIF-Net [26] and WNet [27], are also considered. To ensure the fairness of the experimental results, all comparative models are retrained on identical hardware, strictly adhering to the parameter settings outlined in the original papers.

C. Experimental Setup

1) Evaluation Metrics: The performance evaluation of our proposed method is based on four widely recognized metrics in change detection: precision (Pre), recall (Rec), $F1$-score ($F1$), and intersection over union (IoU). These metrics provide a quantitative measure of the differences between the predicted results and ground truth changes. The formulas for these metrics are as follows:

$$\text{Pre} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{15}$$

$$\text{Rec} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{16}$$

$$F1 = \frac{2 \times \text{Pre} \times \text{Rec}}{\text{Pre} + \text{Rec}} \tag{17}$$

$$\text{IoU} = \frac{\text{TP}}{\text{TP} + \text{FP} + \text{FN}} \tag{18}$$

where TP, FP, and FN represent the number of true positives, false positives, and false negatives, respectively.

2) Implementation Details: For both the training and testing phases, we utilized dual NVIDIA RTX 3090 GPUs, each boasting 24 GB of memory. The training employed the AdamW optimizer with an initial learning rate of 0.0005 and a weight decay of 0.01, utilizing a cosine annealing strategy for learning rate adjustment. Furthermore, during data loading, several data augmentation techniques were applied, including random cropping, flipping, color jittering, and Gaussian noise introduction, to enhance the robustness of the model and prevent overfitting.

D. Comparing Experiments

1) Quantitative Evaluation: Table I presents the comparative quantitative evaluation results of our proposed method against SOTA methods on three public datasets. Scores highlighted in bold and underlined indicate the first and second place in each column, respectively.

For the LEVIR-CD dataset, we observe that the early fusion network FC-EF exhibits the lowest $F1$ and IoU scores at 86.03% and 75.49%, respectively. This outcome suggests that Siamese networks, such as FC-Siam-conc and FC-Siam-diff, outperform early fusion networks in change feature extraction, as evidenced by their significantly superior scores. Furthermore, the inclusion of attention mechanisms in SNUNet, DMINet, and GAS-Net has markedly enhanced detection capabilities, with DMINet and GAS-Net excelling in Pre and Rec scores, respectively. The integration of CNN and transformer technologies in ICIF-Net and WNet shows marginal improvements over the transformer-only BIT method across all metrics. Our proposed method achieves first place in key metrics $F1$ and IoU. It improves by 0.84% and 1.42% compared to the second place GAS-Net.

For the WHU-CD dataset, both the early fusion network FC-EF and the Siamese networks FC-Siam-conc and FC-Siam-diff, based on the UNet architecture, exhibit lower

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TABLE I
COMPARISON OF THE RESULTS WITH OTHER SOTA CHANGE DETECTION METHODS ON THE LEVIR-CD, WHU-CD, AND SYSU-CD DATASETS

Methods	LEVIR-CD				WHU-CD				SYSU-CD			
	Pre(%)	Rec(%)	F1(%)	IoU(%)	Pre(%)	Rec(%)	F1(%)	IoU(%)	Pre(%)	Rec(%)	F1(%)	IoU(%)
FC-EF (2018)	84.60	87.52	86.03	75.49	69.26	77.80	73.28	57.83	70.71	76.48	73.48	58.08
FC-Siam-conc (2018)	89.06	89.23	89.14	80.41	61.05	86.66	71.64	55.81	76.64	77.92	77.27	62.97
FC-Siam-diff (2018)	89.77	88.07	88.91	80.04	75.73	80.29	77.94	63.85	**86.09**	55.70	67.64	51.10
SNUNet (2021)	90.80	89.10	89.94	81.73	85.01	87.29	86.13	75.64	80.01	74.74	77.30	63.00
DMINet (2023)	**92.54**	88.89	90.66	82.92	79.07	89.52	83.97	72.37	85.56	75.64	80.29	67.08
GAS-Net (2023)	90.55	**91.34**	90.94	83.39	87.35	89.88	88.60	79.53	79.32	76.50	77.88	63.78
BIT (2021)	91.00	88.77	89.87	81.60	83.95	86.51	85.21	74.23	80.15	74.93	77.45	63.20
ICIF-Net (2022)	91.40	88.89	90.13	82.03	86.83	85.55	86.18	75.72	76.39	76.99	76.70	62.20
WNet (2023)	91.53	89.29	90.40	82.48	94.44	84.30	89.08	80.31	80.56	79.06	79.81	66.39
Ours	92.46	91.10	**91.78**	**84.81**	**94.79**	**90.97**	**92.84**	**86.64**	83.89	**80.14**	**81.97**	**69.45**

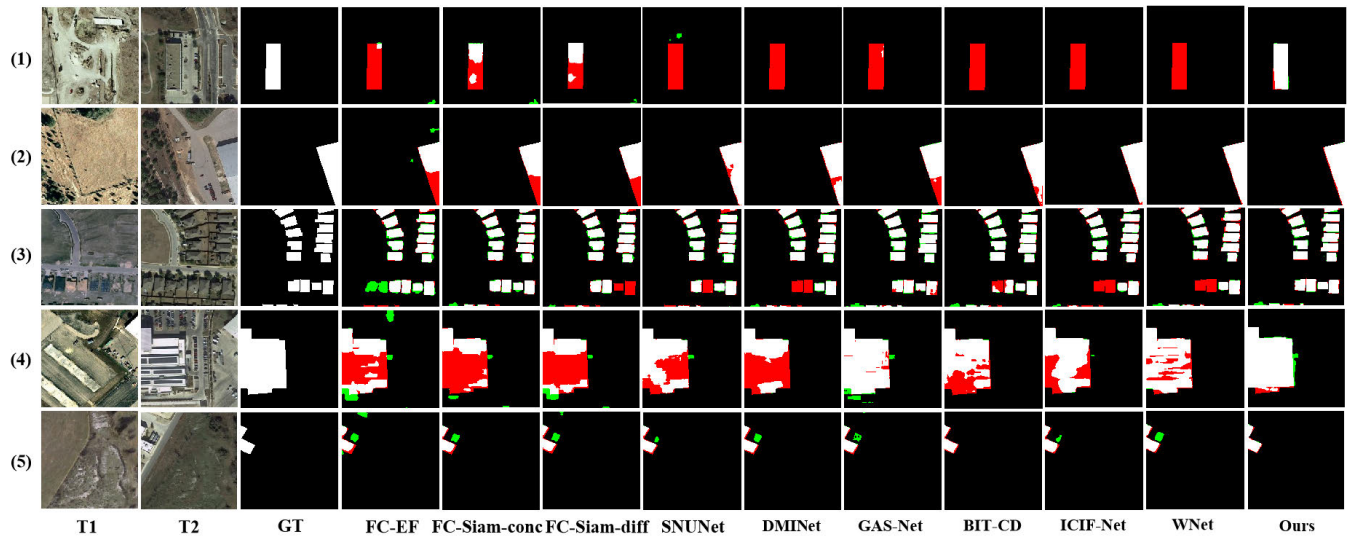


Fig. 7. Visual comparisons of different methods on the LEVIR-CD dataset. Red represents FN, green represents FP, white represents TP (i.e., "changed"), and black represents TN (i.e., "unchanged").

performance across all evaluation metrics compared to other methods. Attention-enhanced CNN methods, notably GAS-Net, underscore the value of attentional mechanisms by ranking high in the Rec score. Among the methods combining CNN and transformer, WNet ranks second in key metrics, including Pre, $F1$, and IoU scores, surpassing other comparative methods in detection performance. However, our proposed method outshines all competitors, ranking first in Pre, Rec, $F1$, and IoU, affirming its superiority with notable margins.

The SYSU-CD dataset encompasses a wider range of change types compared to the LEVIR-CD and WHU-CD datasets, making it more challenging for change detection tasks. Analysis reveals that CNN methods with attention mechanisms generally surpass pure CNN approaches. Furthermore, the fusion of CNN and transformer techniques slightly edges out other models, suggesting that their combined local–global feature modeling optimally captures change information. Nonetheless, our method emerges as the leader in Rec, $F1$, and IoU metrics, underscoring its efficiency and dominance in handling diverse change detection scenarios.

2) Qualitative Evaluation: For qualitative assessment through visualization, we randomly selected five pairs of remote sensing images from the LEVIR-CD, WHU-CD, and SYSU-CD datasets and displayed the comparative results of all methods in Figs. 7–9. In these visualizations, FNs are marked in red, FPs in green, TPs in white, and true negatives (TNs) in black.

Fig. 7 presents the visualization results of various methods on the LEVIR-CD dataset. Specifically, Fig. 7(1)–(4) demonstrates that all compared methods generally exhibit poor integrity in detecting change objects. In addition, Fig. 7(5) illustrates that due to the presence of irrelevant changes, the comparative methods commonly exhibit false detection issues. In contrast, our proposed STFF-GA method performs significantly better than all compared methods across the five image pairs, demonstrating its superiority.

Fig. 8 shows the visualization results of different methods on the WHU-CD dataset. The WHU-CD dataset, with its higher spatial resolution (0.075 m/pixel), provides richer scene details. However, it also brings more disturbing factors such

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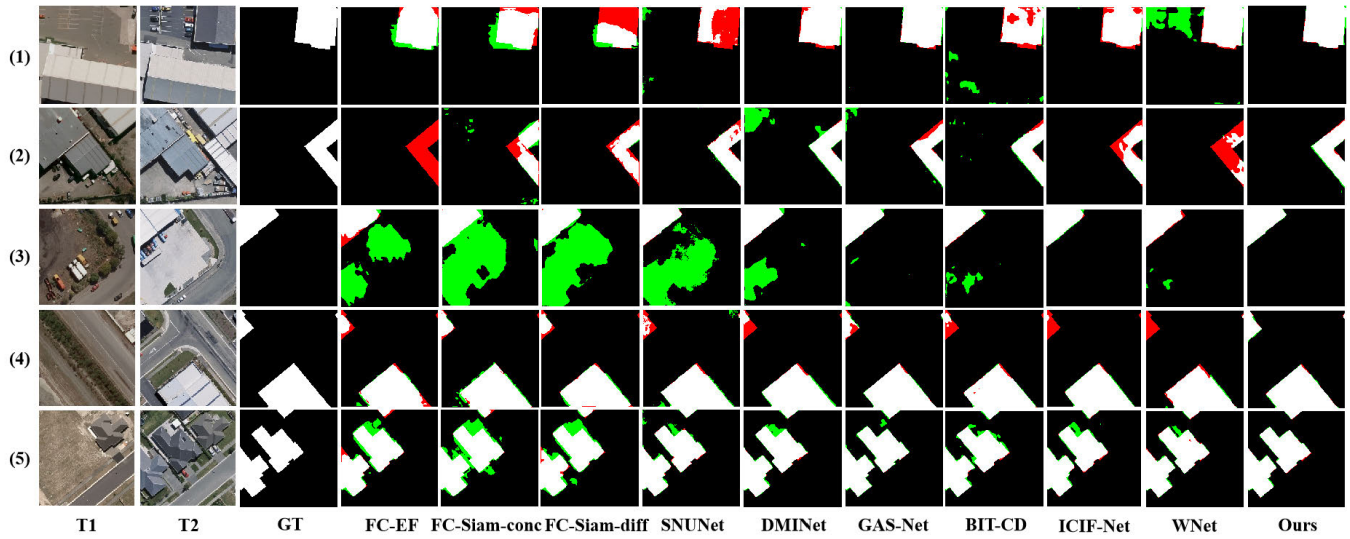


Fig. 8. Visual comparisons of different methods on the WHU-CD dataset. Red represents FN, green represents FP, white represents TP (i.e., "changed"), and black represents TN (i.e., "unchanged").

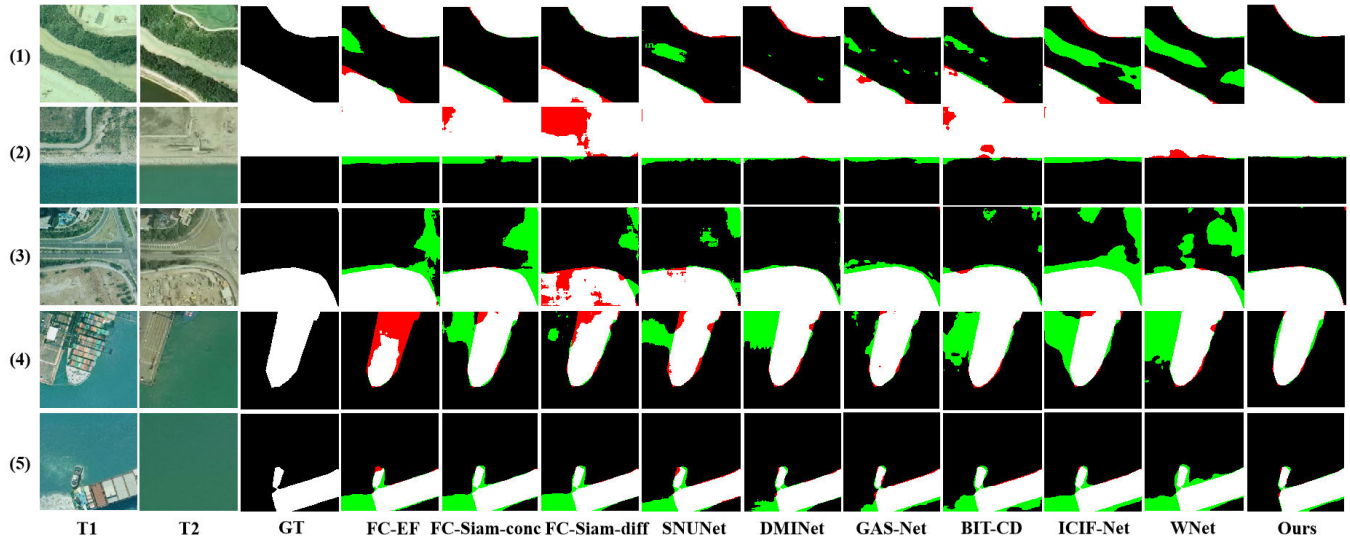


Fig. 9. Visual comparisons of different methods on the SYSU-CD dataset. Red represents FN, green represents FP, white represents TP (i.e., "changed"), and black represents TN (i.e., "unchanged").

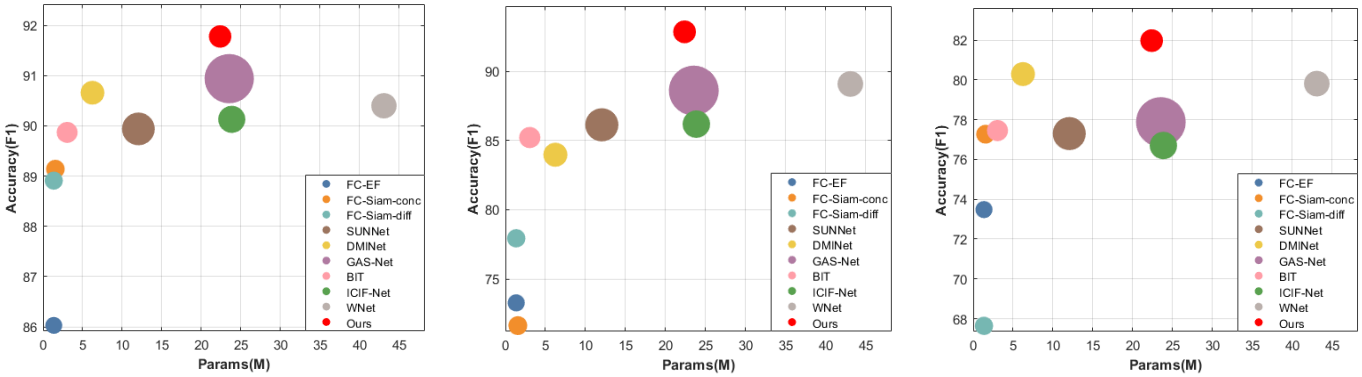
as pseudo-changes. The results reveal that Fig. 8(1) for BIT and WNet, Fig. 8(2) for DMINet, and almost all methods in Fig. 8(3) and (5) exhibit numerous false detections due to pseudo-changes. In contrast, our proposed method can more accurately locate the change object and filter the interference of irrelevant noise, thus achieving accurate and complete change detection.

Fig. 9 presents the visualization results of different methods on the SYSU-CD dataset. Compared to the LEVIR-CD and WHU-CD datasets, the SYSU-CD dataset exhibits a higher rate of false detections. Two main factors contribute to this increased error rate: first, inconsistencies in color within the same region caused by changes in lighting or seasons [as shown in Fig. 9(1) and (2)]. Second, the interference from noninteresting change objects [as shown in Fig. 9(3)–(5)], which complicates accurate classification. Although our proposed method shows a few false detections at the edges of the change regions, its results are very close to the ground truth labels, demonstrating overall satisfactory performance.

3) Comparison of Efficiency: Beyond the quantitative and qualitative evaluation metrics previously discussed, the efficiency of the models is also an important evaluation criterion. Efficiency is typically gauged by two metrics: the number of model parameters (Params) and the computational cost in floating-point operations (FLOPs). In our experiments, the image size employed is 256 × 256 pixels.

The comparison of performance, Params, and FLOPs of different models on the LEVIR-CD, WHU-CD, and SYSU-CD datasets is illustrated in Fig. 10. The size of the circles is directly proportional to the FLOPs. Observing Fig. 10, it can be seen that models such as FC-EF, FC-Siam-conc, and FC-Siam-diff, characterized by simpler feature extraction networks, register lower Params and FLOPs, which correlates with their comparatively modest detection capabilities. Conversely, models like SNUNet, DMINet, and GAS-Net, which incorporate attention mechanisms or advanced feature extraction modules, report enhanced detection outcomes at the expense of increased Params and FLOPs. Notably, DMINet

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(a) (b) (c)

Fig. 10. Comparison of model performance, Params, and FLOPs of different methods on LEVIR-CD, WHU-CD, and SYSU-CD datasets. The area of the circle is proportional to the size of the FLOPs. (a) LEVIR-CD dataset. (b) WHU-CD dataset. (c) SYSU-CD dataset.

TABLE II

QUANTITATIVE PERFORMANCE COMPARISON OF ABLATION STUDY RESULTS ON DIFFERENT DATASETS

Datasets	Model	STFF	GA (w/o DFG)	GA (w/ DFG)	F1(%)	IoU(%)
LEVIR-CD	Baseline	-	-	-	90.73	83.04
	Network1	✓	-	-	91.06	83.59
	Network2	-	-	✓	91.48	84.30
	Network3	✓	✓	-	91.41	84.18
	STFF-GA	✓	-	✓	**91.78**	**84.81**
WHU-CD	Baseline	-	-	-	89.96	81.75
	Network1	✓	-	-	92.16	85.46
	Network2	-	-	✓	92.08	85.33
	Network3	✓	✓	-	92.49	86.04
	STFF-GA	✓	-	✓	**92.84**	**86.64**
SYSU-CD	Baseline	-	-	-	78.82	65.04
	Network1	✓	-	-	80.99	68.05
	Network2	-	-	✓	80.56	67.44
	Network3	✓	✓	-	81.33	68.54
	STFF-GA	✓	-	✓	**81.97**	**69.45**

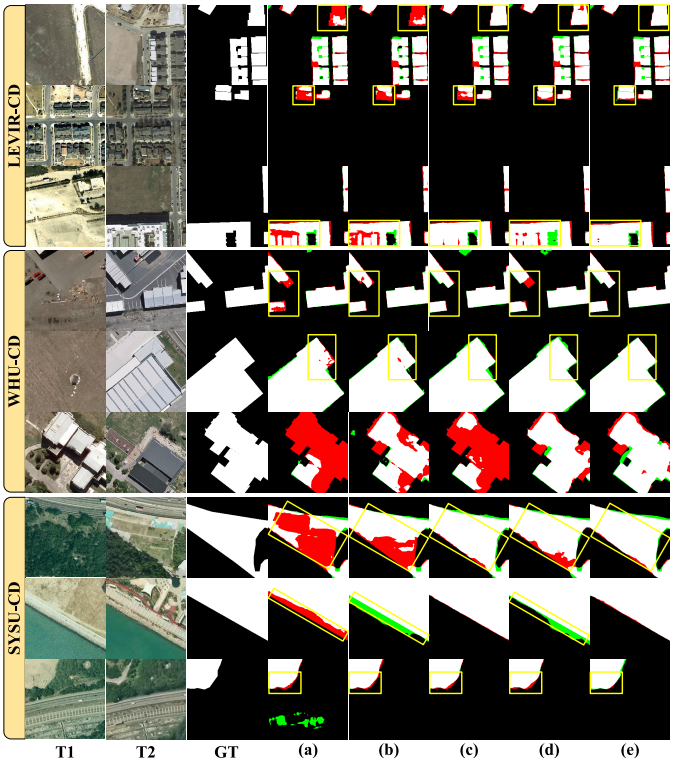


Fig. 11. Visualization results of ablation study on three different datasets. (a) Baseline. (b) Network1. (c) Network2. (d) Network3. (e) STFF-GA. Yellow boxes highlight areas with noticeable detection errors.

stands out for achieving commendable detection performance with relatively low Params. Moreover, ICIF-Net and WNet, which blend CNN and transformer technologies, demonstrate strong change detection capabilities, although WNet exhibits the highest Params. The BIT model, with its lower Params and FLOPs, also delivers impressive detection performance. Compared to these methods, our STFF-GA model exhibits fewer Params than GAS-Net, ICIF-Net, and WNet, and lesser FLOPs compared to SNUNet, DMINet, ICIF-Net, and WNet. Significantly, STFF-GA outperforms all $F1$ metrics in comparative experiments, underscoring its ability to balance superior detection accuracy with efficiency.

E. Ablation Experiments

1) Ablation of the STFF Module and GA Module: To validate the efficacy of the proposed STFF module and GA module, we conducted ablation experiments on three publicly available CD datasets. The experiments included a Baseline (without any additional modules), Network1 (add the STFF module), Network2 (add the GA module with DFG), Network3 (add the STFF and GA modules without DFG), and STFF-GA (add all modules). Specific experimental results are shown in Table II, where w/o denotes "without" and w/ denotes "with," respectively. In addition, the visualization results of the ablation study are shown in Fig. 11.

a) Effectiveness of the STFF module: The STFF module aims to facilitate the interactive fusion of bi-temporal features and effectively capture temporal change features at different levels, thereby enhancing the representation performance of change information. As indicated in Table II, integrating

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TABLE III
QUANTITATIVE COMPARISON RESULTS OF THE THREE DIFFERENT BI-TEMPORAL FEATURE FUSION METHODS

Temporal fusion	Params (M)	FLOPs (G)	LEVIR-CD F1(%)	WHU-CD F1(%)	SYSU-CD F1(%)
Cat(Baseline)	12.28	9.29	90.73	89.96	78.82
Sub	11.73	7.03	89.49	90.34	80.27
STFF	12.28	8.22	**91.06**	**92.16**	**80.99**

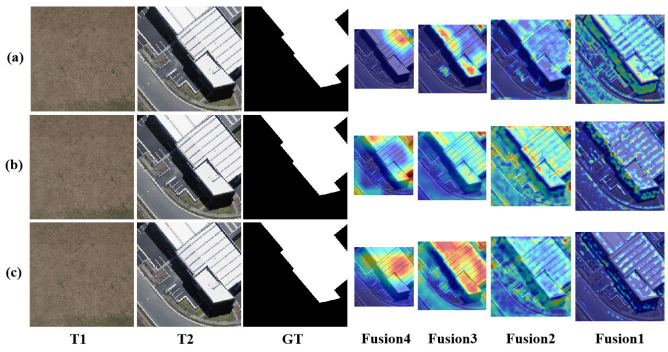


Fig. 12. Visualization heat maps of Cat, Sub, and STFF after fusion of different feature layers. (a) Cat. (b) Sub. (c) STFF. Hotter colors indicate regions with a higher probability of change.

the STFF module into the Baseline network (Network1) results in notable improvements in the $F1$ metric by 0.33%, 2.2%, and 2.17% on the LEVIR-CD, WHU-CD, and SYSU-CD datasets, respectively. Fig. 11(a) and (b) illustrates the detection results of the Baseline and the Baseline with the added STFF module, respectively. It is evident that the incorporation of the STFF module significantly reduces missed detections. The quantitative and qualitative ablation results demonstrate the effectiveness of the STFF module.

To further underscore the effectiveness of the STFF module, we contrasted it against traditional bi-temporal feature fusion techniques, specifically channel-wise concatenation (Cat) and element-wise subtraction (Sub). Replacing the bi-temporal feature fusion method of the Baseline with Cat, Sub, and STFF modules for comparative analysis revealed the fusion method of Baseline aligns with Cat. The comparative outcomes, detailed in Table III, demonstrate that the STFF module surpasses both Cat and Sub methods in the $F1$ and IoU metrics across all datasets. Specifically, the STFF module leads by 0.33%, 1.82%, and 0.72% in $F1$ score compared to the second-best method across the datasets, thus confirming its superior performance. The STFF module maintains nearly the same number of parameters as the Cat method while producing significantly lower FLOPs overhead, further demonstrating its efficiency.

Furthermore, to provide a more intuitive understanding of the advantages of our proposed STFF module, we visualized the heat maps resulting from Cat, Sub, and STFF fusion at different feature layers, as shown in Fig. 12. Note that in Fig. 12, hotter colors indicate regions with a higher probability of change. Fusion1 and Fusion2 represent the fusion of shallow features, which contain more edge and texture information. Fusion3 and Fusion4 represent the fusion of deep features, which contain more semantic change information and are crucial for accurately reconstructing change objects. Observing Fusion3 and Fusion4, it is evident that the heat maps from Cat and Sub fusions show incomplete change detection. In contrast, the proposed STFF module accurately detects most change areas with higher detection probabilities, effectively demonstrating its superior performance in capturing spatio-temporal change features and enhancing the representation of change information.

b) Effectiveness of the GA module: The GA module is designed to guide the aggregation of multilevel semantic information, reduce the interference of pseudo-changes and other noise, and facilitate the comprehensive reconstruction of change objects. To ascertain the effectiveness of the GA module, it was incorporated into the Baseline method (Network2). As shown in Table II, across the LEVIR-CD, WHU-CD, and SYSU-CD datasets, the $F1$ scores of Network 2 improved by 0.83%, 2.12%, and 2.69%, respectively, compared to Baseline, indicating the efficacy of the GA module. As shown in Fig. 11(a) and (c), introducing the GA module into the Baseline effectively mitigates false detections caused by pseudo-changes and other noises and enhances the complete reconstruction of change objects.

Subsequently, to verify the critical role of DFG mapping in locating change objects, we conducted ablation studies using the STFF-GA framework without the DFG module (Network3). Compared to Network3, the STFF-GA model achieved $F1$ score improvements of 0.37%, 0.35%, and 0.64% on the LEVIR-CD, WHU-CD, and SYSU-CD datasets, respectively. Fig. 11(d) and (e) shows that the inclusion of DFG mapping significantly reduces both false and missed detection, aligning with the quantitative findings presented in Table II. In addition, we visualized the decoding processes of STFF-GA (with DFG) and STFF-GA (without DFG), as shown in Fig. 13. The first row illustrates the decoding process of STFF-GA (without DFG), while the second row shows the decoding process of STFF-GA (with DFG). The visualization results indicate that STFF-GA (without DFG) has difficulty in distinguishing pseudo-changes caused by noninteresting changes, frequently misidentifying them as real changes. In contrast, our proposed STFF-GA (with DFG) model, guided by the DFG mapping, accurately locates change objects and filters the interference of pseudo-changes and other noise. This effectively demonstrates the critical role of the DFG mapping within the STFF-GA model.

Furthermore, to confirm the advantages of the DFG module, we introduce two recently proposed change-guidance modules, named CGM1 [43] and CGM2 [56] for comparison. We embedded CGM1, CGM2, and DFG into the STFF-GA framework for comparative analysis. Table IV presents the complexity and performance comparison of these different combinations on the LEVIR-CD, WHU-CD, and SYSU-CD datasets.

Observing Table IV, it is evident that the STFF-GA incorporating the DFG module significantly outperforms the comparative methods in detection performance across all three datasets while maintaining the lowest FLOPs. CGM1 and

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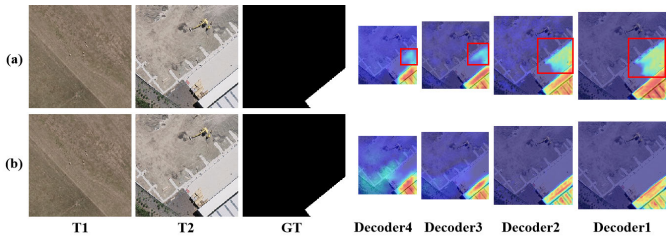


Fig. 13. Visualization heat maps of the decoding process in STFF-GA with and without the DFG module. (a) STFF-GA without DFG module. (b) STFF-GA with DFG module. Red boxes represent pseudo-variable interferences.

TABLE IV
PERFORMANCE COMPARISON OF USING DIFFERENT GUIDANCE MODULES IN STFF-GA

Model	Params (M)	FLOPs (G)	LEVIR-CD F1(%)	WHU-CD F1(%)	SYSU-CD F1(%)
CGM1 [43]	28.18	43.81	91.61	92.50	81.78
CGM2 [56]	20.66	12.99	91.44	92.39	81.63
DFG	22.37	11.67	**91.78**	**92.84**	**81.97**

CGM2 generate change guidance maps through convolution operations using multilevel change features and the deepest change features, respectively. However, CGM1 does not account for pseudo-changes and other background noise in shallow features, and directly concatenating multilevel change features without processing may limit the accuracy of the change guidance map. On the other hand, CGM2 utilizes the deepest high-level features, which despite containing advanced semantic information, undergo multiple convolutions and downsampling, potentially resulting in the loss of detailed information. In addition, these two methods have difficulty in capturing global information over long distances through convolution operations alone. Different from the above methods, our designed DFG module utilizes high-level semantic information at two different scales from the deepest layers. This high-level semantic information is less sensitive to pseudo-changes and other noise, allowing for a more accurate representation of change objects. In addition, the two different scales of high-level features provide richer change information and details. The self-attention mechanism is then employed to capture global information representations of the change features. Therefore, the DFG mapping can more accurately guide the aggregation of multilevel semantic information, correct the location of change objects, and filter out pseudo-changes and other background noise.

2) Ablation Study for the Structural Composition of Different DFG Modules: To validate the superiority of our proposed DFG module combination, we conducted ablation experiments on deep features F^3_{STFF}, F^4_{STFF}, and F^5_{STFF} to identify the most effective DFG network architecture. The configurations of different DFG modules explored in the ablation study are detailed in Table V.

Table VI displays the results of the ablation study on the DFG network with varied architectures. Initially, a single deep feature is employed for the ablation study. The analyses

TABLE V
NETWORK STRUCTURE COMPOSITION OF THE DFG MODULE ABLATION STUDY

Model	Network structure
Base model	STFF-GA(GA module without DFG)
DFG-3	Base model + DFG(F^3_{STFF})
DFG-4	Base model + DFG(F^4_{STFF})
DFG-5	Base model + DFG(F^5_{STFF})
DFG-3-4	Base model + DFG(F^3_{STFF} and F^4_{STFF})
DFG-3-5	Base model + DFG(F^3_{STFF} and F^5_{STFF})
DFG-4-5	Base model + DFG(F^4_{STFF} and F^4_{STFF})
DFG-3-4-5	Base model + DFG(F^3_{STFF}, F^4_{STFF}, and F^5_{STFF})

TABLE VI
QUANTITATIVE PERFORMANCE COMPARISON FOR ABLATION STUDY OF DFG MODULES WITH DIFFERENT STRUCTURES

Methods	LEVIR-CD		WHU-CD		SYSU-CD	
	F1(%)	IoU(%)	F1(%)	IoU(%)	F1(%)	IoU(%)
Base model	91.41	84.18	92.49	86.04	81.33	68.54
DFG-3	91.50	84.33	92.43	85.93	81.35	68.56
DFG-4	91.41	84.17	91.85	84.93	81.53	68.82
DFG-5	91.48	84.30	92.04	85.25	81.52	68.80
DFG-3-4	91.57	84.45	92.50	86.05	81.43	68.67
DFG-3-5	91.51	84.36	92.31	85.72	<u>81.85</u>	<u>69.29</u>
DFG-4-5	**91.78**	**84.81**	**92.84**	**86.64**	**81.97**	**69.45**
DFG-3-4-5	<u>91.60</u>	<u>84.51</u>	<u>92.63</u>	<u>86.28</u>	81.54	68.83

of DFG-3, DFG-4, and DFG-5 reveal that the detection performance of these configurations is inferior to the Base model on the WHU-CD dataset, suggesting that reliance on single-layer deep features may lead to overly abstracted change information or loss of pertinent details, thereby reducing detection accuracy. In the subsequent phase of the study, which involved dual-layer deep features, the detection results on the three datasets mostly surpassed those of methods relying solely on single-depth features and the Base model. Notably, the DFG-4-5 achieves the highest scores for the key metrics $F1$ and IoU across all three datasets. In addition, the detection performance of DFG-3-4-5 is marginally inferior to that of DFG-4-5, necessitating additional parameters and computational resources. This decrease in performance can be attributed to the potential amalgamation of irrelevant or noisy features from three levels, negatively impacting DFG mapping efficiency. The complexity of change objects within the SYSU-CD dataset further exacerbates this negative impact. Therefore, extensive ablation experiments have fully demonstrated the superiority of our employing F^4_{STFF} and F^5_{STFF} to construct DFG modules.

3) Ablation Study of Deeply Supervised for DFG Mapping: Our designed loss function comprises the DSL for DFG mapping and the model output loss. By applying deep supervision to the DFG mapping, the DFG mapping can more closely

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TABLE VII
ABLATION STUDY OF DEEPLY SUPERVISED FOR DFG MAPPING

Datasets	Model	F1(%)	IoU(%)
LEVIR-CD	STFF-GA(w/o DSL)	91.52	84.36
	STFF-GA(w/ DSL)	**91.78**	**84.81**
WHU-CD	STFF-GA(w/o DSL)	92.64	86.29
	STFF-GA(w/ DSL)	**92.84**	**86.64**
SYSU-CD	STFF-GA(w/o DSL)	81.87	69.31
	STFF-GA(w/ DSL)	**81.97**	**69.45**

approximate the ground truth. This provides more accurate change object location information for the aggregation of multilevel semantic information, promoting the precise reconstruction of change features. To verify the effectiveness of applying DSL on the DFG mapping, we conducted ablation studies of STFF-GA (w/ DSL) and STFF-GA (w/o DSL) on the LEVIR-CD, WHU-CD, and SYSU-CD datasets. Table VII presents the results of these ablation studies. Here, w/o represents "without" and w/ represents "with." The results indicate that adding DFG DSL improved the detection performance of the model by 0.26%, 0.20%, and 0.10% on the LEVIR-CD, WHU-CD, and SYSU-CD datasets, respectively, confirming the effectiveness of the DFG mapping DSL.

V. CONCLUSION

In this article, we propose an innovative STFF-GA network for change detection tasks, focusing on addressing the challenges of inadequate spatio-temporal change representation and insufficient multilevel semantic information aggregation. To this end, we designed the low-parameter, low-computation STFF module, which employs a strategy of splitting, interaction, and fusion to effectively capture bi-temporal change features and enhance the representation of change information. Subsequently, we introduced a GA module that integrates the DFG module and multiscale convolution operations to effectively aggregate multilevel change information, ensuring complete reconstruction of change features. Extensive experiments conducted on three benchmark datasets demonstrate the effectiveness and robustness of the proposed STFF-GA network. Also, ablation studies were conducted, which sufficiently validated the effectiveness of each module in our proposed model.

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